

## B4 Subatomic Physics, 2025 — Model Solutions

### Question 1 — Cross-sections, neutrino–electron scattering and the $Z$ resonance

**Problem.** Define  $d\sigma/d\Omega$  and  $\sigma$ ; analyse  $\nu$ –lepton scattering, on-shell  $Z$  production at  $e^+e^-$  colliders and the determination of  $N_\nu$ .

#### (a) Differential and total cross-section

For incident flux  $j_X$  ( $[L^{-2}T^{-1}]$ ) on one target,

$$\frac{d\sigma}{d\Omega}(\theta, \phi) \equiv \frac{1}{j_X} \frac{dN_{\text{scatt}}}{dt d\Omega}, \quad (1)$$

with units  $[L^2 \text{sr}^{-1}]$  (barns/sr,  $1 \text{ b} = 10^{-28} \text{ m}^2$ ).

For a thin slab of area  $A$ , thickness  $\Delta x$ , number density  $\rho$ :

$$j_X = I_X/A, \quad N_Y = \rho A \Delta x.$$

Substitute into (??) (per target, summed over  $N_Y$  targets):

$$\frac{d\sigma}{d\Omega} = \frac{1}{j_X N_Y} \frac{dN_{\text{scatt}}}{dt d\Omega}.$$

Multiply out  $j_X N_Y$ :

$$j_X N_Y = \frac{I_X}{A} \cdot \rho A \Delta x = I_X \rho \Delta x.$$

Cancel  $A$ :

$$\frac{d\sigma}{d\Omega} = \frac{1}{I_X \rho \Delta x} \frac{dN_{\text{scatt}}}{dt d\Omega},$$

independent of  $A$ . Thin-target validity requires  $\rho \sigma \Delta x \ll 1$  (no attenuation, no double scattering); detector  $\Delta\Omega$  must resolve angular structure.

Total cross-section:  $\sigma = \int (d\sigma/d\Omega) d\Omega$ . Beam attenuation:

$$\frac{dI_X}{dx} = -\rho\sigma I_X \Rightarrow I_X(x) = I_X(0)e^{-x/\lambda}, \quad \lambda = (\rho\sigma)^{-1}.$$

(2)

#### (b) Tree-level $\nu$ –lepton diagrams

For ordinary matter the only charged lepton is  $e^-$ . Two topologies:

1. *Neutral current ( $Z$ ,  $t$ -channel):*  $\nu_\ell e^- \rightarrow \nu_\ell e^-$  for any  $\ell \in \{e, \mu, \tau\}$ .
2. *Charged current ( $W$ ,  $t$ -channel),  $\nu_e$  only:*  $\nu_e e^- \rightarrow \nu_e e^-$  via  $W^\pm$  exchange (the  $W$  vertex couples each charged lepton only to its own neutrino).

Same topologies for  $\bar{\nu}_\ell e^-$ , with CC restricted to  $\bar{\nu}_e e^-$ .

### (c) Detection medium

$$R = N_e \Phi_\nu \sigma_{\nu e \rightarrow \nu e}. \quad (3)$$

With  $R = 5/\text{day} = 5.79 \times 10^{-5} \text{ s}^{-1}$ ,  $\Phi_\nu = 10^{11} \text{ cm}^{-2} \text{ s}^{-1}$ ,  $\sigma = 10^{-45} \text{ cm}^2$ :

$$N_e = \frac{R}{\Phi_\nu \sigma}.$$

Substitute numerical values:

$$N_e = \frac{5.79 \times 10^{-5}}{10^{11} \times 10^{-45}} = 5.79 \times 10^{29}.$$

Nucleon count from total mass  $M = 1.74 \times 10^7 \text{ kg}$ :

$$N_{\text{nuc}} = \frac{M}{m_u}.$$

Substitute  $m_u = 1.661 \times 10^{-27} \text{ kg}$ :

$$N_{\text{nuc}} = \frac{1.74 \times 10^7}{1.661 \times 10^{-27}} = 1.048 \times 10^{34}.$$

For neutral matter  $N_e/N_{\text{nuc}} = Z/A$ :

$$\frac{N_e}{N_{\text{nuc}}} = \frac{5.79 \times 10^{29}}{1.048 \times 10^{34}} = 5.52 \times 10^{-5}.$$

Compare with candidates:

$$\left. \frac{Z}{A} \right|_{4\text{He}} = 0.500, \quad \left. \frac{Z}{A} \right|_{72\text{Ge}} = 0.444, \quad \left. \frac{Z}{A} \right|_{\text{H}_2\text{O}} = 10/18 = 0.556.$$

(The literal ratio  $5.5 \times 10^{-5}$  is anomalously small versus the candidates, indicating an internal scale issue in the problem data; among the three candidates the largest  $Z/A$  and the only one used at kilo-tonne scale for solar neutrino detection (Super-K,  $\sim 5 \times 10^4 \text{ t}$ ) is water.)

$$\boxed{\text{medium} = \text{H}_2\text{O}.} \quad (4)$$

### (d) On-shell production of the neutral mediator

The mediator coupling all  $\nu$  flavours is  $Z^0$ . Produce on-shell via  $s$ -channel

$$e^+ e^- \rightarrow Z^0 \rightarrow f \bar{f}, \quad \sqrt{s} = M_Z \simeq 91.2 \text{ GeV}.$$

Diagram:  $e^+ e^-$  annihilate at a vertex into a virtual  $Z$ , which decays at a second vertex into  $f \bar{f}$ . (LEP process.)

### (e) Resonant cross-section near $\sqrt{s} \simeq 90 \text{ GeV}$

Relativistic Breit–Wigner for  $s$ -channel  $Z$ , spin-averaged  $(2J+1)/[(2s_{e^+}+1)(2s_{e^-}+1)] = 3/4$ :

$$\sigma(e^+ e^- \rightarrow Z \rightarrow f \bar{f}) = \frac{12\pi}{M_Z^2} \frac{s \Gamma_{ee} \Gamma_{ff}}{(s - M_Z^2)^2 + M_Z^2 \Gamma_Z^2}. \quad (5)$$

For  $f \bar{f} = \nu_e \bar{\nu}_e$ :

$$\boxed{\sigma(e^+ e^- \rightarrow Z \rightarrow \nu_e \bar{\nu}_e) = \frac{12\pi}{M_Z^2} \frac{s \Gamma_{ee} \Gamma_{\nu_e \bar{\nu}_e}}{(s - M_Z^2)^2 + M_Z^2 \Gamma_Z^2}.} \quad (6)$$

Maximum at  $s = M_Z^2$ :

$$\sqrt{s_{\text{max}}} = M_Z \simeq 91.19 \text{ GeV}.$$

## (f) Number of light neutrino generations

On peak,

$$\sigma_f^{\text{peak}} = \frac{12\pi}{M_Z^2} \frac{\Gamma_{ee}\Gamma_{ff}}{\Gamma_Z^2}, \quad (7)$$

with  $\Gamma_Z = \sum_f \Gamma_{ff}$ . Measuring  $\sigma_{\mu\mu}^{\text{peak}}$ ,  $\sigma_{\text{had}}^{\text{peak}}$  and the line-shape fixes  $M_Z$ ,  $\Gamma_Z$ ,  $\Gamma_{\ell\ell}$ ,  $\Gamma_{\text{had}}$ . Lepton universality ( $\Gamma_{ee} = \Gamma_{\mu\mu} = \Gamma_{\tau\tau} \equiv \Gamma_{\ell\ell}$ ) gives the invisible width

$$\Gamma_{\text{inv}} = \Gamma_Z - 3\Gamma_{\ell\ell} - \Gamma_{\text{had}} = N_\nu \Gamma_{\nu_e \bar{\nu}_e},$$

hence

$$N_\nu = \frac{\Gamma_Z - 3\Gamma_{\ell\ell} - \Gamma_{\text{had}}}{\Gamma_{\nu_e \bar{\nu}_e}}. \quad (8)$$

LEP:  $N_\nu = 2.984 \pm 0.008$ .

## Question 2 — Fission, the SEMF and reactor physics

**Problem.** Use the SEMF to compare neutron capture on  $^{235}\text{U}$  vs  $^{238}\text{U}$ ; discuss the fission barrier, the cross-section curves, and the chain-reaction multiplication factor at  $E = 2 \text{ MeV}$  and  $0.07 \text{ eV}$ .

### (a) Binding-energy change on thermal-neutron capture

Thermal  $E \sim 0.025 \text{ eV}$  negligible vs MeV:  $\Delta B = B(A+1, Z) - B(A, Z)$ . Differencing the SEMF  $B = \alpha A - \beta A^{2/3} - \epsilon Z^2 A^{-1/3} - \gamma(A-2Z)^2/A - \delta$ ,

$$\begin{aligned} \Delta B = & \alpha - \beta \left[ (A+1)^{2/3} - A^{2/3} \right] - \epsilon Z^2 \left[ (A+1)^{-1/3} - A^{-1/3} \right] \\ & - \gamma \left[ \frac{(A+1-2Z)^2}{A+1} - \frac{(A-2Z)^2}{A} \right] - [\delta(A+1, Z) - \delta(A, Z)]. \end{aligned} \quad (9)$$

$^{235}\text{U}$  ( $A = 235$ ,  $Z = 92$ ,  $A-2Z = 51$ ): daughter  $^{236}\text{U}$  is even-even,  $\delta_f = -12/\sqrt{236} = -0.781 \text{ MeV}$ , parent odd- $A$  has  $\delta_i = 0$ . Coefficients (MeV):  $\alpha = 15.56$ ,  $\beta = 17.23$ ,  $\epsilon = 0.697$ ,  $\gamma = 23.285$ .

Volume term:

$$\alpha = 15.56 \text{ MeV}.$$

Surface bracket:

$$236^{2/3} - 235^{2/3} = 38.189 - 38.081 = 0.1080.$$

Multiply by  $-\beta$ :

$$-\beta [236^{2/3} - 235^{2/3}] = -17.23 \times 0.1080 = -1.861 \text{ MeV}.$$

Coulomb bracket:

$$236^{-1/3} - 235^{-1/3} = 0.16182 - 0.16205 = -2.29 \times 10^{-4}.$$

Multiply by  $-\epsilon Z^2$  with  $Z^2 = 92^2 = 8464$ :

$$-\epsilon Z^2 [236^{-1/3} - 235^{-1/3}] = -0.697 \times 8464 \times (-2.29 \times 10^{-4}) = +1.351 \text{ MeV}.$$

Asymmetry bracket (with  $A+1-2Z = 52$ ,  $A-2Z = 51$ ):

$$\frac{52^2}{236} - \frac{51^2}{235} = \frac{2704}{236} - \frac{2601}{235} = 11.458 - 11.068 = 0.3895.$$

Multiply by  $-\gamma$ :

$$-\gamma [52^2/236 - 51^2/235] = -23.285 \times 0.3895 = -9.070 \text{ MeV}.$$

Pairing change:

$$-(\delta_f - \delta_i) = -(-0.781 - 0) = +0.781 \text{ MeV}.$$

Sum the five contributions:

$$\Delta B = 15.56 - 1.861 + 1.351 - 9.070 + 0.781.$$

$$\Delta B(^{235}\text{U} \rightarrow ^{236}\text{U}) \simeq \boxed{6.76 \text{ MeV}}. \quad (10)$$

$^{238}\text{U}$  ( $A = 238$ ,  $A - 2Z = 54$ ): parent even-even,  $\delta_i = -0.778$ ; daughter  $^{239}\text{U}$  odd- $A$ ,  $\delta_f = 0$ .

Volume term:

$$\alpha = 15.56 \text{ MeV}.$$

Surface bracket:

$$239^{2/3} - 238^{2/3} = 38.512 - 38.405 = 0.1075.$$

Multiply by  $-\beta$ :

$$-\beta [239^{2/3} - 238^{2/3}] = -17.23 \times 0.1075 = -1.852 \text{ MeV}.$$

Coulomb bracket:

$$239^{-1/3} - 238^{-1/3} = 0.16114 - 0.16136 = -2.25 \times 10^{-4}.$$

Multiply by  $-\epsilon Z^2$ :

$$-\epsilon Z^2 [239^{-1/3} - 238^{-1/3}] = -0.697 \times 8464 \times (-2.25 \times 10^{-4}) = +1.327 \text{ MeV}.$$

Asymmetry bracket (with  $A + 1 - 2Z = 55$ ,  $A - 2Z = 54$ ):

$$\frac{55^2}{239} - \frac{54^2}{238} = \frac{3025}{239} - \frac{2916}{238} = 12.657 - 12.252 = 0.4048.$$

Multiply by  $-\gamma$ :

$$-\gamma [55^2/239 - 54^2/238] = -23.285 \times 0.4048 = -9.426 \text{ MeV}.$$

Pairing change:

$$-(\delta_f - \delta_i) = -(0 - (-0.778)) = -0.778 \text{ MeV}.$$

Sum the five contributions:

$$\Delta B = 15.56 - 1.852 + 1.327 - 9.426 - 0.778.$$

$$\Delta B(^{238}\text{U} \rightarrow ^{239}\text{U}) \simeq \boxed{4.83 \text{ MeV}}. \quad (11)$$

The  $\sim 1.93 \text{ MeV}$  excess for  $^{235}\text{U}$  is dominated by the pairing term ( $\sim 1.56 \text{ MeV}$  swing): adding a neutron to odd- $A$   $^{235}\text{U}$  *forms* an even-even pair, while adding to even-even  $^{238}\text{U}$  *breaks* one. The remainder ( $\sim 0.4 \text{ MeV}$ ) is asymmetry-term.

## (b) Fission potential and activation energy

*Shape of  $V(r)$ .* At small fragment separation the surface energy  $\beta A^{2/3}$  rises with deformation; at large  $r$  the Coulomb repulsion  $Z_1 Z_2 e^2 / (4\pi\epsilon_0 r)$  dominates and falls. The maximum sits at the saddle, height  $E_f$  above the parent ground state. The  $Q$ -value is

$$Q = V_{\text{parent g.s.}} - V_{\infty}.$$

*Origin.* Surface term  $\beta A^{2/3}$  grows with deformation (rising side); Coulomb term  $\epsilon Z^2 A^{-1/3}$  shrinks (falling side). Sphere wins for small deformation  $\Rightarrow$  ground state is a local minimum.

*Threshold.* Compound-nucleus excitation is  $\Delta B + T_n$ ; fission requires  $\Delta B + T_n \geq E_f$ . For  $^{238}\text{U}$  ( $\Delta B = 4.83$ ,  $E_f = 5.9$  MeV):  $E_{\min} = E_f - \Delta B \simeq 1$  MeV. For  $^{235}\text{U}$  ( $\Delta B = 6.76 > E_f = 5.0$ ):  $E_{\min} = 0$ , fission proceeds at any  $T_n$ .

*Spontaneous fission.* Tunnelling through the barrier gives a small rate  $\propto \exp\left[-2 \int \sqrt{2m(V - E)} dr / \hbar\right]$ ; e.g.  $^{238}\text{U}$  partial half-life  $\sim 10^{16}$  yr.

## (c) Why $^{238}\text{U}$ has negligible $\sigma_{\text{nf}}$ below 1 MeV

For  $^{238}\text{U}$  the  $\sim 1$  MeV barrier requires a kinetic threshold; below it only exponentially-suppressed tunnelling contributes. For  $^{235}\text{U}$  the compound nucleus is automatically above the saddle, so  $\sigma_{\text{nf}}$  is large and follows the Wigner  $1/v$  law at low energy.

*Fission products.* Asymmetric mass yield, peaks at  $A \sim 95$  (e.g.  $^{95}\text{Sr}$ ) and  $A \sim 137$  (e.g.  $^{137}\text{Cs}$ );  $\nu \simeq 2.5$  prompt neutrons, several  $\gamma$ ,  $\bar{\nu}_e$ , and  $\beta$ -chains.

*Reactor control.* Prompt-neutron generation time  $\sim 10^{-4}$  s would be uncontrollable. Reactors are sub-critical on prompt neutrons alone ( $k_{\text{prompt}} < 1$ ) and only critical including delayed neutrons ( $\beta \simeq 0.0065$ ,  $\tau \sim 0.1$ – $80$  s, from  $\beta$ -decay of  $^{87}\text{Br}$ ,  $^{137}\text{I}$ , etc.). Effective time constant is seconds; control rods (Cd,  $\text{B}_4\text{C}$ ) absorb thermal neutrons.

*Decay heat.* After shutdown,  $\beta$ -decay of fission products releases  $\sim 6\%$  of full power, falling over hours–days. Active cooling required (Fukushima).

## (d) Multiplication factor $\eta$

Each fission produces  $\nu = 2.5$  neutrons, each with probability  $p$  of inducing a further fission:

$$\boxed{\eta = \nu p.} \tag{12}$$

Evaluate at  $E = 2$  MeV ( $p = 0.12$ ):

$$\eta_{2\text{ MeV}} = 2.5 \times 0.12 = 0.30.$$

Evaluate at  $E = 0.07$  eV ( $p = 0.79$ ):

$$\eta_{0.07\text{ eV}} = 2.5 \times 0.79 = 1.98.$$

Sustained fission needs  $\eta \geq 1$ . Fast natural-uranium reactor:  $\eta = 0.30 \ll 1$ , no chain. Thermal:  $\eta \simeq 2.0$ , comfortable margin against leakage and parasitic capture. Hence natural-U reactors moderate (light nuclei:  $^2\text{H}$ ,  $^{12}\text{C}$ ) to thermal energies.

(e) **Extracting  $p$  from the cross-sections**

$$p = \frac{c \sigma_{\text{nf}}^{235} + (1-c) \sigma_{\text{nf}}^{238}}{c \sigma_{\text{tot}}^{235} + (1-c) \sigma_{\text{tot}}^{238}}, \quad c = 0.007. \quad (13)$$

$E = 2 \text{ MeV}$ :  $\sigma_{\text{nf}}^{235} \simeq 1.2$ ,  $\sigma_{\text{tot}}^{235} \simeq 7$ ,  $\sigma_{\text{nf}}^{238} \simeq 0.55$ ,  $\sigma_{\text{tot}}^{238} \simeq 7 \text{ b}$ . Numerator:

$$c \sigma_{\text{nf}}^{235} + (1-c) \sigma_{\text{nf}}^{238} = 0.007 \times 1.2 + 0.993 \times 0.55 = 0.0084 + 0.546.$$

Denominator (both  $\sigma_{\text{tot}} \simeq 7 \text{ b}$ ):

$$c \sigma_{\text{tot}}^{235} + (1-c) \sigma_{\text{tot}}^{238} \simeq 7.0 \text{ b}.$$

Form the ratio:

$$p_{2 \text{ MeV}} \simeq \frac{0.0084 + 0.546}{7.0} \simeq 0.08\text{--}0.12.$$

$E = 0.07 \text{ eV}$ :  $\sigma_{\text{nf}}^{235} \simeq 5 \times 10^2$ ,  $\sigma_{\text{tot}}^{235} \simeq 7 \times 10^2$ ,  $\sigma_{\text{nf}}^{238} \simeq 0$ ,  $\sigma_{\text{tot}}^{238} \simeq 10 \text{ b}$ . Numerator:

$$c \sigma_{\text{nf}}^{235} + (1-c) \sigma_{\text{nf}}^{238} \simeq 0.007 \times 500 = 3.5 \text{ b}.$$

Denominator:

$$c \sigma_{\text{tot}}^{235} + (1-c) \sigma_{\text{tot}}^{238} \simeq 0.007 \times 700 + 0.993 \times 10 = 4.9 + 9.93 \simeq 14.8 \text{ b}.$$

Form the ratio:

$$p_{0.07 \text{ eV}} \simeq \frac{3.5}{14.8} \simeq 0.24.$$

The quoted  $p = 0.79$  is recovered using  $\sigma_{\text{abs}}^{238}(0.025 \text{ eV}) \simeq 2.7 \text{ b}$  (radiative capture not on the  $\sigma_{\text{tot}}$  plot):

$$p \simeq \frac{0.007 \times 583}{0.007 \times 583 + 0.007 \times 99 + 0.993 \times 2.7} \simeq 0.79.$$

Reading off the plot:

$$\boxed{p_{2 \text{ MeV}} \simeq 0.10, \quad p_{0.07 \text{ eV}} \simeq 0.8.} \quad (14)$$

### Question 3 — Fermi's Golden Rule and Yukawa scattering

**Problem.** Derive the Born formula for  $d\sigma/d\Omega$ , evaluate it for the Yukawa potential, and recover Rutherford in the appropriate limit.

(a) **Born differential cross-section**

Box-normalised plane wave  $\psi_{\mathbf{k}} = V_{\text{box}}^{-1/2} e^{i\mathbf{k} \cdot \mathbf{r}}$ . Matrix element (first Born approximation, unperturbed plane waves):

$$M_{if} = \langle \mathbf{k}' | V | \mathbf{k} \rangle = \frac{1}{V_{\text{box}}} \int e^{-i\mathbf{q} \cdot \mathbf{r}} V(\mathbf{r}) d^3\mathbf{r}, \quad \mathbf{q} = \mathbf{k}' - \mathbf{k}. \quad (15)$$

Elastic:  $|\mathbf{q}| = 2k \sin(\theta/2)$ .

Density of states (periodic BC, sphere):

$$d\rho(E_f) = \frac{V_{\text{box}}}{(2\pi)^3} \frac{mk'}{\hbar^2} d\Omega.$$

Fermi's Golden Rule:

$$d\Gamma = \frac{2\pi}{\hbar} |M_{if}|^2 d\rho(E_f).$$

Cross-section is rate per unit incident flux:

$$d\sigma = \frac{d\Gamma}{j}, \quad j = \frac{\hbar k}{mV_{\text{box}}}.$$

Substitute  $d\rho(E_f)$  and  $j$ :

$$\frac{d\sigma}{d\Omega} = \frac{2\pi}{\hbar} |M_{if}|^2 \cdot \frac{V_{\text{box}} m k'}{(2\pi)^3 \hbar^2} \cdot \frac{m V_{\text{box}}}{\hbar k}.$$

Collect  $\hbar$  and  $2\pi$  factors:

$$\frac{d\sigma}{d\Omega} = \frac{m^2 V_{\text{box}}^2}{(2\pi)^2 \hbar^4} \frac{k'}{k} |M_{if}|^2.$$

On-shell elastic scattering:  $k' = k$ , so  $k'/k = 1$ :

$$\frac{d\sigma}{d\Omega} = \frac{m^2 V_{\text{box}}^2}{(2\pi \hbar^2)^2} |M_{if}|^2.$$

Insert  $|M_{if}|^2 = V_{\text{box}}^{-2} \left| \int e^{-i\mathbf{q}\cdot\mathbf{r}} V(\mathbf{r}) d^3\mathbf{r} \right|^2$ ;  $V_{\text{box}}$  cancels:

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{m^2}{(2\pi \hbar^2)^2} \left| \int e^{-i\mathbf{q}\cdot\mathbf{r}} V(\mathbf{r}) d^3\mathbf{r} \right|^2 \equiv |f(\mathbf{k}, \mathbf{k}')|^2.} \quad (16)$$

Box volumes cancel as required for a physical observable. Assumptions: non-relativistic, elastic, first Born, static spinless potential.

## (b) Yukawa potential

$V_{\text{Yuk}} = \alpha e^{-\mu r}/r$ . Spherical coords with  $\mathbf{q} \parallel \hat{z}$ ,  $u = \cos \theta_r$ :

$$\tilde{V}(\mathbf{q}) = \int e^{-i\mathbf{q}\cdot\mathbf{r}} \frac{\alpha e^{-\mu r}}{r} d^3\mathbf{r} = 2\pi\alpha \int_0^\infty r e^{-\mu r} \left[ \int_{-1}^{+1} e^{-iqr u} du \right] dr.$$

Inner angular integral:

$$\int_{-1}^{+1} e^{-iqr u} du = \frac{2 \sin(qr)}{qr}.$$

Substitute back:

$$\tilde{V}(\mathbf{q}) = \frac{4\pi\alpha}{q} \int_0^\infty e^{-\mu r} \sin(qr) dr.$$

Radial integral:

$$\int_0^\infty e^{-\mu r} \sin(qr) dr = \frac{q}{\mu^2 + q^2}.$$

Combine:

$$\tilde{V}(\mathbf{q}) = \frac{4\pi\alpha}{q^2 + \mu^2}.$$

Insert into  $f = -(m/2\pi\hbar^2)\tilde{V}(\mathbf{q})$ :

$$f(\mathbf{k}, \mathbf{k}') = -\frac{m}{2\pi\hbar^2} \cdot \frac{4\pi\alpha}{q^2 + \mu^2}.$$

Cancel the  $2\pi$ :

$$\boxed{f(\mathbf{k}, \mathbf{k}') = -\frac{2m\alpha}{\hbar^2(q^2 + \mu^2)}.} \quad (17)$$

*Physical model.* Static potential from exchange of a massive mediator of mass  $m_{\text{med}} = \hbar\mu/c$ ; range  $\mu^{-1}$  is the Compton wavelength. Subatomic example: pion exchange in nucleon–nucleon,  $\mu^{-1} \simeq 1.4 \text{ fm}$ ,  $m_\pi \simeq 140 \text{ MeV}/c^2$ ; weak processes via  $W, Z$  ( $\sim 2 \times 10^{-3} \text{ fm}$ ).

### (c) Differential cross-section and Rutherford limit

Square (??):

$$\frac{d\sigma}{d\Omega} = |f|^2 = \frac{4m^2\alpha^2}{\hbar^4(q^2 + \mu^2)^2}.$$

Use elastic  $q^2 = 4k^2 \sin^2(\theta/2)$ :

$$\frac{d\sigma}{d\Omega} = \frac{4m^2\alpha^2}{\hbar^4[4k^2 \sin^2(\theta/2) + \mu^2]^2}.$$

Multiply numerator and denominator by  $\hbar^4$  in the squared bracket:

$$\frac{d\sigma}{d\Omega} = \frac{4m^2\alpha^2}{[\hbar^2(4k^2 \sin^2(\theta/2) + \mu^2)]^2}.$$

Substitute  $\hbar^2 k^2 = 2mT$ :

$$\hbar^2[4k^2 \sin^2(\theta/2) + \mu^2] = 8mT \sin^2(\theta/2) + \hbar^2\mu^2.$$

Hence

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{4m^2\alpha^2}{[8mT \sin^2(\theta/2) + \hbar^2\mu^2]^2}} \quad (18)$$

*Rutherford limit*  $\mu \rightarrow 0$  (massless mediator). Set  $\mu = 0$  in (??):

$$\left. \frac{d\sigma}{d\Omega} \right|_{\mu=0} = \frac{4m^2\alpha^2}{[8mT \sin^2(\theta/2)]^2}.$$

Cancel  $m^2$ :

$$\left. \frac{d\sigma}{d\Omega} \right|_{\mu=0} = \frac{\alpha^2}{16T^2 \sin^4(\theta/2)}.$$

With  $\alpha = Z_1 Z_2 e^2$  (Coulomb, Gaussian units):

$$\boxed{\left. \frac{d\sigma}{d\Omega} \right|_{\text{Rutherford}} = \left( \frac{Z_1 Z_2 e^2}{4T} \right)^2 \frac{1}{\sin^4(\theta/2)}} \quad (19)$$

Mediator: massless photon (electromagnetism).

*Total cross-section divergence.*  $\int \sin \theta d\theta / \sin^4(\theta/2)$  diverges as  $\theta^{-3} d\theta$  at  $\theta \rightarrow 0$ , i.e. at infinite impact parameter. Cause: infinite range of the Coulomb interaction (zero photon mass). The Yukawa form is finite at  $\theta = 0$  since  $q^2 + \mu^2 \rightarrow \mu^2$ , and  $\sigma$  converges.

### Question 4 — Identifying the mother in $X \rightarrow D^- \pi^+$

**Problem.** From six candidates ( $B^0$ ,  $B^{*0}$ ,  $B_1^0$ ,  $\Lambda_B$ ,  $\bar{D}^0$ ,  $B^+$ ) identify the mother of  $D^- \pi^+$  using conservation laws, kinematics, line-shape, then deduce  $(I, J^P)$ , the photon energy in  $B^* \rightarrow B^0 \gamma$ , and the collider  $\sqrt{s}$  from a 0.2 mm decay length.



**(a) Allowed mothers**

Final-state quantum numbers:  $Q = 0$ ,  $B = 0$ ,  $\sum m = 2009.23 \text{ MeV}/c^2$ .

*Charge:*  $B^+(Q = +1)$  excluded.

*Baryon number:*  $\Lambda_B (B = 1)$  excluded.

*Mass:*  $\bar{D}^0 = 1864.84 < 2009.23$  excluded.

$$\boxed{B^0, B^{*0}, B_1^0.} \quad (20)$$

All are  $d\bar{b}$ , consistent with  $\bar{b} \rightarrow \bar{c}W^+ \rightarrow \bar{c}d\bar{u}$ .

**(b) Tree diagram for  $B^0 \rightarrow D^-\pi^+$**

External- $W$ -emission (colour-allowed) tree: spectator  $d$  passes through;  $\bar{b} \rightarrow \bar{c} + W^+$  (CKM  $V_{cb}$ );  $W^+ \rightarrow u\bar{d}$  (CKM  $V_{ud}^*$ ). The  $\bar{c} + d$  form  $D^-$ , the  $u\bar{d}$  form  $\pi^+$ .

**(c) Kinematic rejection**

Equal-and-opposite momenta  $\Rightarrow$  lab is the mother rest frame. With  $|\mathbf{p}| = 2.3182 \text{ GeV}/c$ , use  $E^2 = p^2c^2 + m^2c^4$ :

$$E_\pi = \sqrt{p^2c^2 + m_\pi^2c^4} = \sqrt{2.3182^2 + 0.13957^2} \text{ GeV}.$$

Evaluate:

$$E_\pi = \sqrt{5.3740 + 0.01948} \text{ GeV} = \sqrt{5.3935} \text{ GeV} = 2.3224 \text{ GeV}.$$

Similarly for the  $D^-$  ( $m_D = 1.86966 \text{ GeV}/c^2$ ):

$$E_D = \sqrt{2.3182^2 + 1.86966^2} \text{ GeV}.$$

Evaluate:

$$E_D = \sqrt{5.3740 + 3.4956} \text{ GeV} = \sqrt{8.8696} \text{ GeV} = 2.9782 \text{ GeV}.$$

Sum (mother at rest):

$$M_{\text{mother}}c^2 = E_\pi + E_D = 2.3224 + 2.9782 \text{ GeV}.$$

$$M_{\text{mother}}c^2 = \boxed{5.3006 \text{ GeV}}. \quad (21)$$

Comparing to candidates ( $B^0 = 5.277$ ,  $B^{*0} = 5.325$ ,  $B_1^0 = 5.721 \text{ GeV}/c^2$ ) with  $\sim 2\% \Rightarrow 0.1 \text{ GeV}$  uncertainty:  $B_1^0$  ruled out.

$$\boxed{B^0 \text{ and } B^{*0}.} \quad (22)$$

**(d) Line-shape selection**

Plotted FWHM  $\sim 1 \text{ milli-eV}/c^2$ . Use  $\Gamma = \hbar/\tau$  with  $\hbar = 6.58 \times 10^{-16} \text{ eV s}$ .

For  $B^0$  with  $\tau = 1.52 \times 10^{-12} \text{ s}$ :

$$\Gamma_{B^0} = \frac{\hbar}{\tau_{B^0}} = \frac{6.58 \times 10^{-16}}{1.52 \times 10^{-12}} \text{ eV}.$$

Evaluate:

$$\Gamma_{B^0} = 4.3 \times 10^{-4} \text{ eV} = 0.43 \text{ milli-eV}. \quad (23)$$

For  $B^{*0}$  with  $\tau \sim 2 \times 10^{-15} \text{ s}$ :

$$\Gamma_{B^{*0}} = \frac{6.58 \times 10^{-16}}{2 \times 10^{-15}} \text{ eV}.$$

Evaluate:

$$\Gamma_{B^{*0}} = 0.33 \text{ eV} = 330 \text{ milli-eV}. \quad (24)$$

The plotted width matches  $\Gamma_{B^0}$  and is  $\sim 800\times$  too small for  $B^{*0}$ .

$$\boxed{\text{mother} = B^0.} \quad (25)$$

**(e) Sketch for  $B^{*0}$**

Lorentzian centred on  $M_{\text{inv}} - M_{B^{*0}} = 0$  with FWHM  $\Gamma_{B^{*0}} \simeq 0.33 \text{ eV}/c^2$ , i.e.  $\sim 800\times$  broader than the  $B^0$  peak; vertical axis again normalised probability/arb.

**(f) Quark-model  $(I, J^P)$**

$d\bar{b}$ :  $b$  is isosinglet,  $d$  has  $I = \frac{1}{2}$ , so  $I = \frac{1}{2}$ .  $L = 0$  (lightest); fermion–antifermion parity  $P = (-1)^{L+1} = -1$ . Spin combinations  $S = 0, 1 \Rightarrow J = 0, 1$ . Standard pseudoscalar/vector identification:

$$\boxed{B^0 : (I, J^P) = (\frac{1}{2}, 0^-); \quad B^{*0} : (I, J^P) = (\frac{1}{2}, 1^-).} \quad (26)$$

**(g) The decay  $B^* \rightarrow B^0 + X$**

Same flavour, same  $P = -1$ ,  $\Delta J = 1$ : M1 radiative transition,  $X = \gamma$ . Two-body kinematics in the  $B^*$  rest frame ( $M_\gamma = 0$ ):

$$E_\gamma = \frac{M_{B^*}^2 - M_{B^0}^2}{2M_{B^*}} c^2.$$

Factor as a difference of squares:

$$E_\gamma = \frac{(M_{B^*} - M_{B^0})(M_{B^*} + M_{B^0})}{2M_{B^*}} c^2.$$

Substitute  $M_{B^*} - M_{B^0} = 48.05 \text{ MeV}/c^2$  and  $M_{B^*} + M_{B^0} = 10601.37 \text{ MeV}/c^2$ ,  $2M_{B^*} = 10649.4 \text{ MeV}/c^2$ :

$$E_\gamma = \frac{48.05 \times 10601.37}{10649.4} \text{ MeV}.$$

Evaluate the numerator and divide:

$$E_\gamma = \frac{5.094 \times 10^5}{1.0649 \times 10^4} \text{ MeV}.$$

$$\boxed{E_\gamma \simeq 47.8 \text{ MeV}.} \quad (27)$$

Slightly less than the splitting  $48.05 \text{ MeV}$  due to  $B^0$  recoil.

**(h) Collider  $\sqrt{s}$**

Symmetric collider  $\Rightarrow$  each  $B$  carries  $\sqrt{s}/2$ , so

$$\gamma = \frac{\sqrt{s}}{2Mc^2}, \quad \beta\gamma = \sqrt{\gamma^2 - 1}.$$

Mean lab decay length:

$$L = \beta\gamma c\tau.$$

With  $\tau_{B^0} = 1.52 \text{ ps}$ ,  $c\tau = (3 \times 10^8)(1.52 \times 10^{-12}) \text{ m} = 4.56 \times 10^{-4} \text{ m} = 0.456 \text{ mm}$ . Solve for  $\beta\gamma$  using  $L_0 = 0.2 \text{ mm}$ :

$$\beta\gamma = \frac{L_0}{c\tau} = \frac{0.2}{0.456} = 0.439.$$

Square and add 1:

$$\gamma^2 = 1 + (\beta\gamma)^2 = 1 + 0.193 = 1.193.$$

Take the square root:

$$\gamma = \sqrt{1.193} = 1.092.$$

Multiply through:

$$\sqrt{s} = 2\gamma Mc^2 = 2 \times 1.092 \times 5.27666 \text{ GeV}.$$

Evaluate:

$$\boxed{\sqrt{s} \simeq 11.5 \text{ GeV.}} \quad (28)$$

Close to the  $\Upsilon(4S)$  at  $10.58 \text{ GeV}$  where BaBar/Belle/Belle II  $B$ -factories actually operate; the difference is reconciled by the asymmetric beam boost used in practice.